Lab 17 — Capstone: Produce an Executive Finance Analytics Pack

Python Programming for Finance · TGS-2026065049 · LO1–LO7

Scenario

The CFO requests a concise, reproducible evidence pack combining data quality, portfolio exposure, returns, risk and recommendations.

Goal

A self-contained HTML report, charts, curated CSVs, validation log and executive summary.

Tools

Python, Pandas, Matplotlib, HTML

Step-by-step

1. Review the acceptance checklist and supplied portfolio data.

2. Run the data-quality audit and save the reconciliation.

3. Calculate position values, weights and sector exposure.

4. Calculate return, volatility and drawdown indicators.

```
python starter.py
```

5. Generate charts and the HTML executive report.

```
python solution.py
```

6. Run validate_outputs.py and capture PASS evidence.

```
python validate_outputs.py
```

Acceptance criteria

All required files exist, totals reconcile, charts have labels, the HTML report opens locally, and validate_outputs.py prints CAPSTONE PASS.

Evidence

Capture the executed output, validation result, generated file or chart, and a one-sentence finance interpretation.